

Diapositiva 1 - **76 palabras (0,7 minutos) - MATEO**



Simulation of the Place Charles de Gaulle Roundabout, Paris (Arc de Triomphe)

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We are **[DECIR LOS NOMBRES DE LOS INTEGRANTES]**, students at Universidad Tecnológica Nacional, Facultad Regional Rosario—commonly known as UTN **[PRONUNCIARLO EN ESPAÑOL]**—in Argentina.

This study uses AnyLogic to improve traffic at the Place Charles de Gaulle roundabout by comparing intervention scenarios against a baseline of current conditions. This project was developed for our Simulation class, focusing on creating a realistic model to evaluate performance under peak-hour assumptions.

Abstract: Modeling Complex Traffic Flow

The Place Charles de Gaulle roundabout is highly complex due to multiple access/exit points, heterogeneous flows, and priority rules. This study models and simulates the roundabout using AnyLogic to measure and reduce the average time each vehicle spends in the system.

Simulation Focus

AnyLogic model integrating 2D/3D representation, vehicle arrivals, probabilistic routing, and control strategies.

Key Metric

Average time from vehicle arrival at one of the 12 access roads to exit via one of the 12 exit roads.

Control System

Traffic light control by opposing pairs (180 sec cycle: 30 sec green / 150 sec red).

Keywords: Simulation - Arc de Triomphe - Roundabout - Congestion - Peak hours

This roundabout is a complex system due to its twelve access points and entry-priority rules, making intuition-based interventions unreliable.

[SEÑALAR TARJETA: SIMULATION FOCUS]

We built a model to compare scenarios in a consistent environment, where vehicles follow probabilistic routing based on network structure.

[SEÑALAR TARJETA: KEY METRIC]

Our primary metric is the average time-in-system, $\bar{T}$, measuring the delay from arrival to exit. This provides a direct measure of overall delay and allows us to compare scenarios in an easy way.

[SEÑALAR TARJETA: CONTROL SYSTEM]

We test traffic-light regulation using opposing pairs on a 180-second cycle. This setup allows us to isolate the effects of signal control strategies during peak conditions.

Diapositiva 3 - 129 palabras (1,2 minutos) - **MATEO**

The Problem: Congestion at Place Charles de Gaulle

The roundabout connects twelve major avenues, including the iconic Avenue des Champs-Élysées, creating complex vehicular dynamics.

Structural Issues

The roundabout lacks clearly delimited lanes, though it accommodates about eight vehicles in parallel.

Operational Difficulties

Frequent bottlenecks, sudden stops, and minor accidents due to high density and lack of clear lane markings.

Unpredictable Behaviour

Real-world peak hour traffic includes erratic behaviour (e.g., vehicles stopping at 90°, dangerous lane crossing) that is difficult to replicate in simulation.



The problem is recurrent congestion at a junction connecting twelve major avenues, including the Champs-Élysées. High demand and entry priority create difficult-to-predict dynamics. Structurally, the lack of delimited lanes forces drivers to negotiate space, causing the system to shift rapidly from flowing to saturated.

[SEÑALAR IMAGEN SUPERIOR]

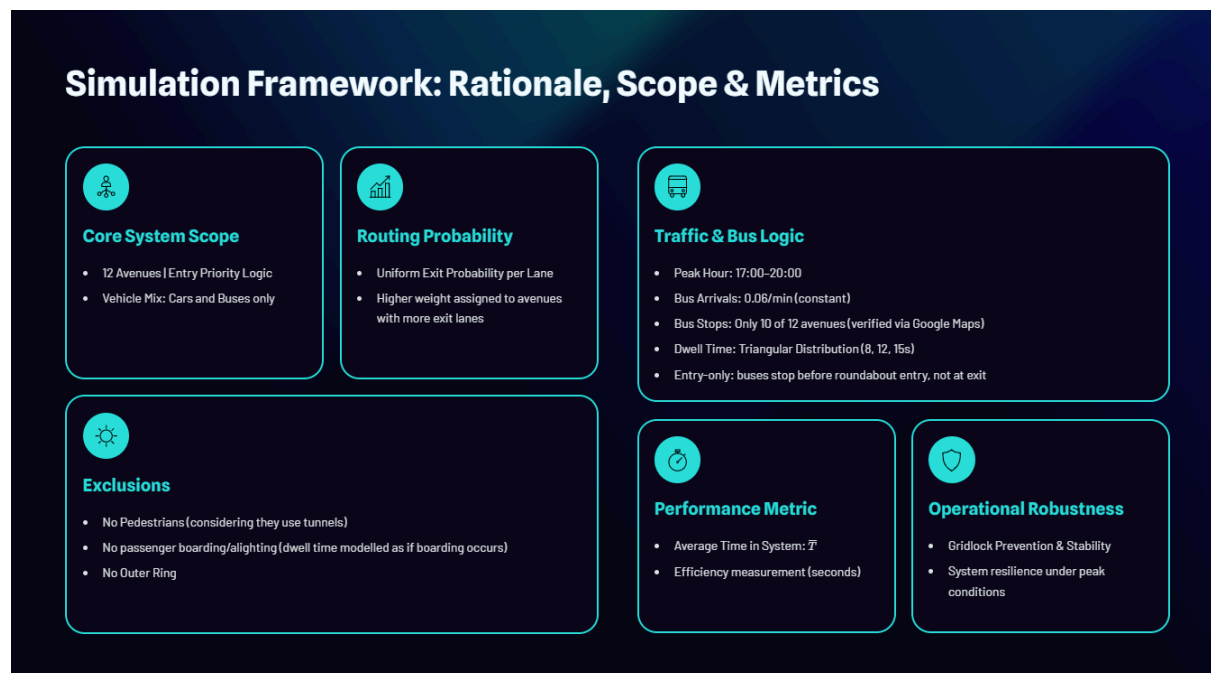
These bottlenecks propagate quickly. The red circles highlight areas where flow becomes constrained and movement stops: vehicles are tightly packed, movement becomes stop-and-go, and a short interruption can create a queue that spreads back to the approaches.

[SEÑALAR IMAGEN INFERIOR]

We also see unpredictable behaviors, like vehicles trapped at 90-degree angles. This occurs when drivers cannot merge out of the inner ring due to jams, creating abrupt stops.

These interactions justify our simulation approach: we aim to identify control strategies that stabilize these patterns.

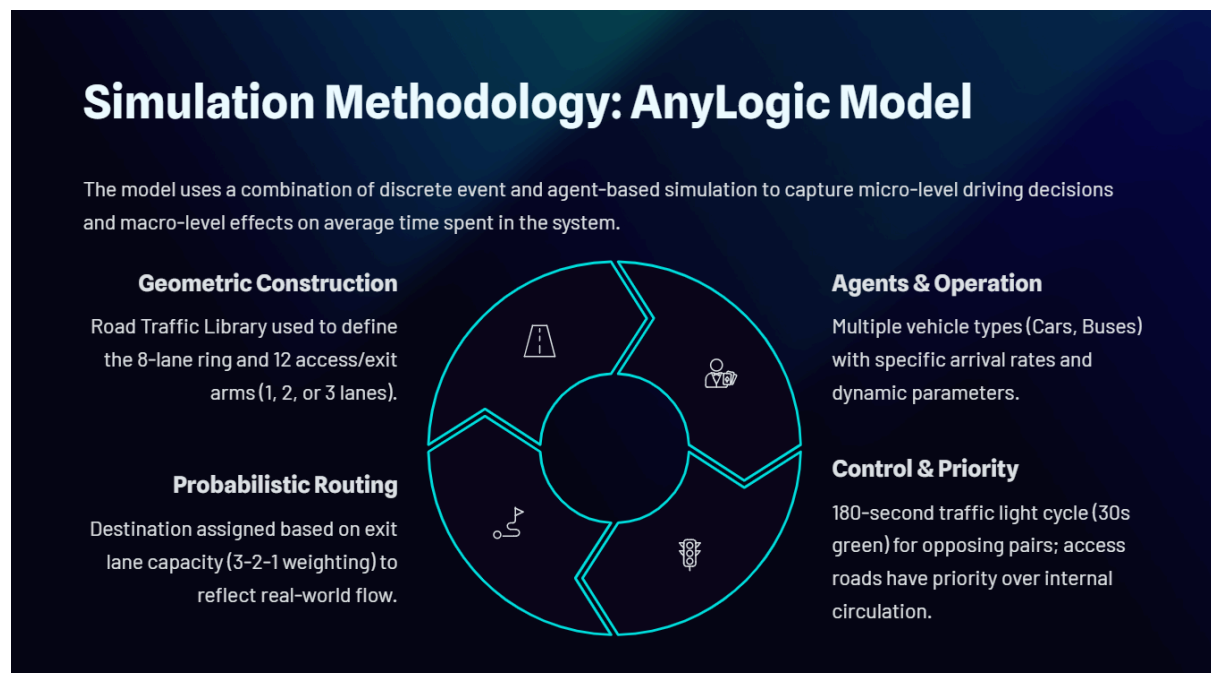
Diapositiva 4 - 144 palabras (1,3 minutos) - ALEXIS



Simulation allows us to test "What-if" scenarios in a risk-free environment, providing data-driven evidence for infrastructure changes. Our scope is defined by four strategic assumptions:

- **Core System:** A car-and-bus-only model of the 12-avenue geometry to isolate signal impact.
- **Routing:** Destinations are assigned based on exit lane capacity, reflecting real-world flow patterns.
- **Traffic & Public Transport:** We focused strictly on the evening peak from 5 PM to 8 PM. Buses arrive at a constant rate of 0.06 per minute—roughly 1 every 17 minutes—from the 10 avenues with bus stops. Dwell times follow a triangular distribution of 8, 12, and 15 seconds.
- **Exclusions:** We excluded pedestrians because they use tunnels and do not interact with vehicle flow.

We evaluate performance via average time-in-system ($\bar{T}$) and Operational Robustness—the system's ability to prevent total saturation.



We used AnyLogic's Road Traffic Library to model the network, including traffic lights and bus stops.

[SEÑALAR ÍCONO: GEOMETRIC CONSTRUCTION]

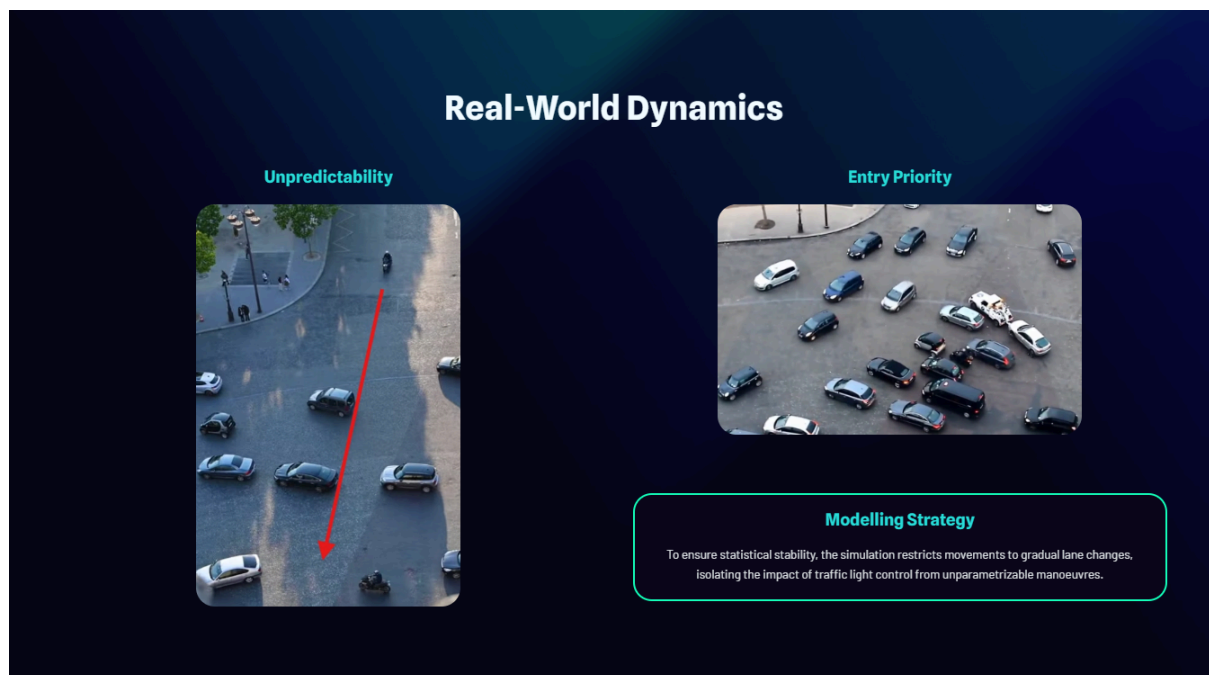
To balance realism and control, we simplified lane usage: vehicles entering from left lanes move to the inner ring, while right-lane entries use the outer lanes.

[SEÑALAR ÍCONO: CONTROL & PRIORITY]

Signals operate on a 180-second cycle, preserving the current "priority-to-the-right" rule. We concluded that expanding the signal control system is far more feasible in practice than trying to reverse this long-standing priority law.

[SEÑALAR ÍCONO: PROBABILISTIC ROUTING]

Destinations are assigned using a 3-2-1 weighting system based on lane capacity. This ensures avenues with 3 lanes attract proportionally higher flow, matching real-world statistics.



Having established our framework, we must now examine the complex real-world dynamics that define this system. The most critical factor is the Entry Priority rule.

[SEÑALAR LADO DERECHO]

Unlike in Argentina, here approaching vehicles have the right of way.

[REPRODUCIR VIDEO DE PRIORIDAD DE ENTRADA]

As you can see in the footage, this rule prevents long avenue queues but increases internal congestion and the risk of gridlock.

[SEÑALAR LADO IZQUIERDO DE LA DIAPOSITIVA: UNPREDICTABILITY]

[REPRODUCIR VIDEO DE LA MOTO CORTANDO CARRILES]

Also, The lack of lanes leads to unpredictable manoeuvres, like vehicles crossing several streams at sharp angles to exit. These behaviors are difficult to parameterize.

enter or

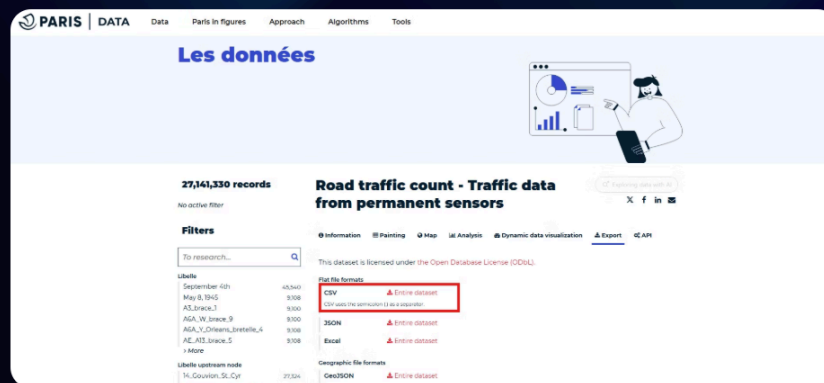
[SEÑALAR CUADRO INFERIOR: MODELLING STRATEGY]

Therefore, our strategy constrains vehicles to gradual lane changes. This stabilizes the model, ensuring performance gains are the result of signal logic, not "stochastic noise".

Diapositiva 7 - 69 palabras (0,6 minutos) - JUAN CRUZ

Data Collection Sources

The detailed traffic flow data used in our simulation, including vehicle counts and arrival rates for each avenue, was sourced from the official opendata.paris.fr portal. This open-source dataset provided the foundational real-world statistics necessary for calibrating our model.



The traffic flow CSV file, aggregated by avenue and time of day, was downloaded from the data portal and provided as input to ChatGPT to obtain the λ values at 100% for each avenue. We will see how these were implemented in the arrival rates table on the next slide.

The traffic flow data used to estimate vehicle arrivals during peak hours was obtained from the official Open Data Paris portal.

[SEÑALAR RECUADRO ROJO: CSV ENTIRE DATASET]

We processed millions of records using ChatGPT to extract specific arrival rates for the twelve approaches.

[SEÑALAR TEXTO INFERIOR DE LA DIAPOSITIVA]

We modelled these arrivals as a Poisson process, using the parameter λ to represent expected inflow per minute. This provided the baseline statistics necessary to calibrate our model for peak-hour conditions.

Diapositiva 8 - 113 palabras (1 minuto) - JUAN CRUZ

Data Calibration & Operational Stress Test			
Avenue Name	Access Lanes (Entry to Roundabout)	Estimated Peak Demand (veh./min)	Max. Sustainable Load (veh./min)
		λ at 100%	λ at 35%
Avenue de la Grande-Armée	3	15	5.25
Avenue Marceau	2	13	4.55
Avenue de Wagram	2	13	4.55
Avenue Kléber	2	12	4.20
Avenue des Champs-Élysées	3	11	3.85
Avenue Foch	2	11	3.85
Avenue Mac-Mahon	2	9	3.15
Avenue Carnot	2	8	2.80
Avenue Hoche	1	7	2.45
Avenue de Friedland	2	7	2.45
Avenue d'Iéna	1	6	2.10
Avenue Victor-Hugo	1	5	1.75

It is challenging to reproduce all erratic behaviors in simulation.

[SEÑALAR COLUMNA: ESTIMATED PEAK DEMAND (λ AT 100%)]

When using 100% of real-world demand, the model reached system collapse immediately—a state of total saturation. By collapse, we mean a total saturation of the junction's capacity, where traffic queues grow indefinitely and the roundabout becomes completely blocked, preventing any further vehicle movement.

We conducted a stress test to find the system's ceiling. At 30% demand, the simulation oscillates between stability and collapse, mirroring real peak conditions.

[SEÑALAR COLUMNA: MAX. SUSTAINABLE LOAD (λ AT 35%)]

We used this threshold to test our dual-signal strategies, finding they could sustain up to 35% demand while maintaining stability. This improvement in the breakdown threshold is a key indicator of robustness.

Diapositiva 9 - 109 palabras (1 minuto) - JUAN CRUZ



Vehicle Routing and Bus Dynamics

Vehicle entry into the roundabout is restricted based on the access road lane, preventing immediate congestion and conflicts observed in initial tests.

Lane Restriction Logic	Bus Agent Implementation
Vehicles in the outer access lane are restricted to the outer roundabout lanes. Vehicles in the inner access lane are restricted to the inner roundabout lanes. This prevents vehicles on the same avenue from blocking each other when entering the roundabout, which caused significant delays and blockages.	Buses were introduced to add complexity, including a simulated stop time before entering the roundabout. Stop time follows a triangular distribution (8, 12, 15 seconds), simulating passenger boarding/alighting at 10 of the 12 access roads.

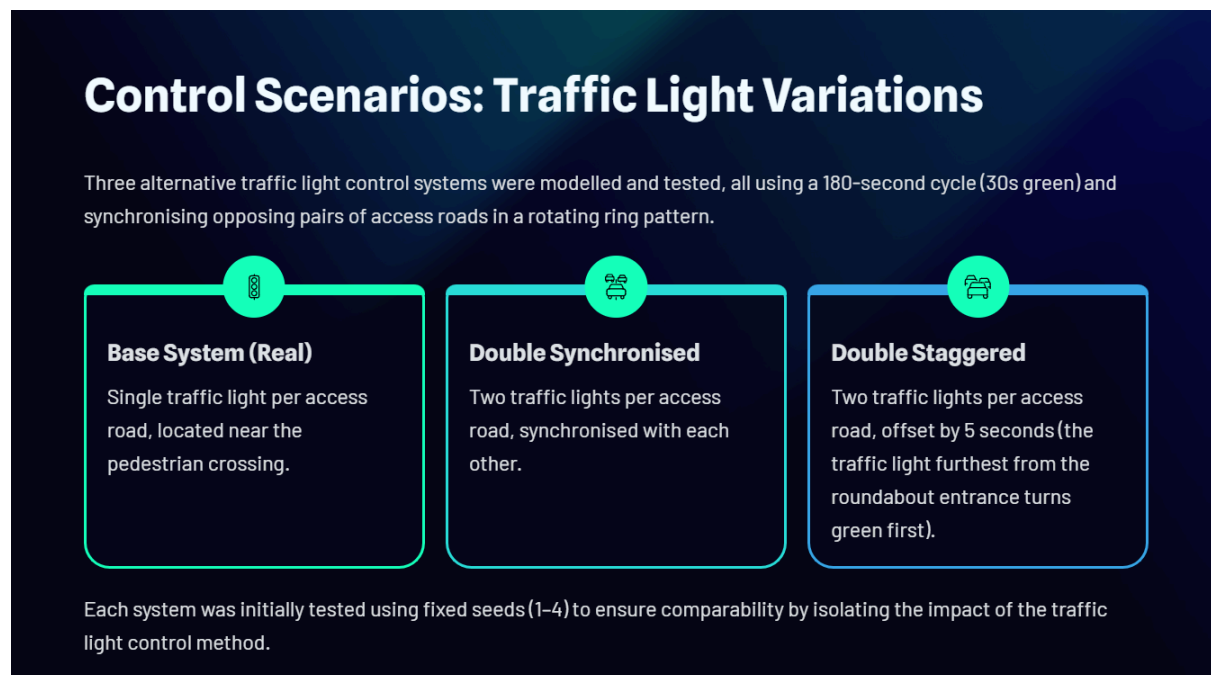
Two implementation details were vital for stability.

[SEÑALAR COLUMNA: LANE RESTRICTION LOGIC]

First, We introduced lane-access constraints at entry: This means that vehicles approaching from the outer access lane are restricted to the outer roundabout lanes, while those in the inner lane are directed to the inner part of the ring. This prevents vehicles from blocking each other at entry points, reducing artificial congestion.

[SEÑALAR COLUMNA: BUS AGENT IMPLEMENTATION]

In addition, Buses stop at 10 of the 12 access roads. Their dwell time follows a triangular distribution with (8, 12, 15s), simulating passenger boarding and alighting. These choices capture real-world dynamics without overcomplicating the model, ensuring results are driven by our control strategies.



We defined a set of traffic-light control scenarios as potential enhancements to the baseline configuration. The purpose here is to isolate the impact of different signal arrangements. By keeping all other variables constant, we can ensure that the improvements we see are driven strictly by the control method we are testing.

[SEÑALAR BLOQUE IZQUIERDO: BASE SYSTEM (REAL)]

Our benchmark is the Base System, which represents the current real-world configuration with a single traffic light per avenue.

[SEÑALAR BLOQUE CENTRAL: DOUBLE SYNCHRONISED]

Next, we evaluated the Double Synchronised System. This configuration uses two lights per road operating in total synchrony to enforce a coordinated stopping pattern at the entry.

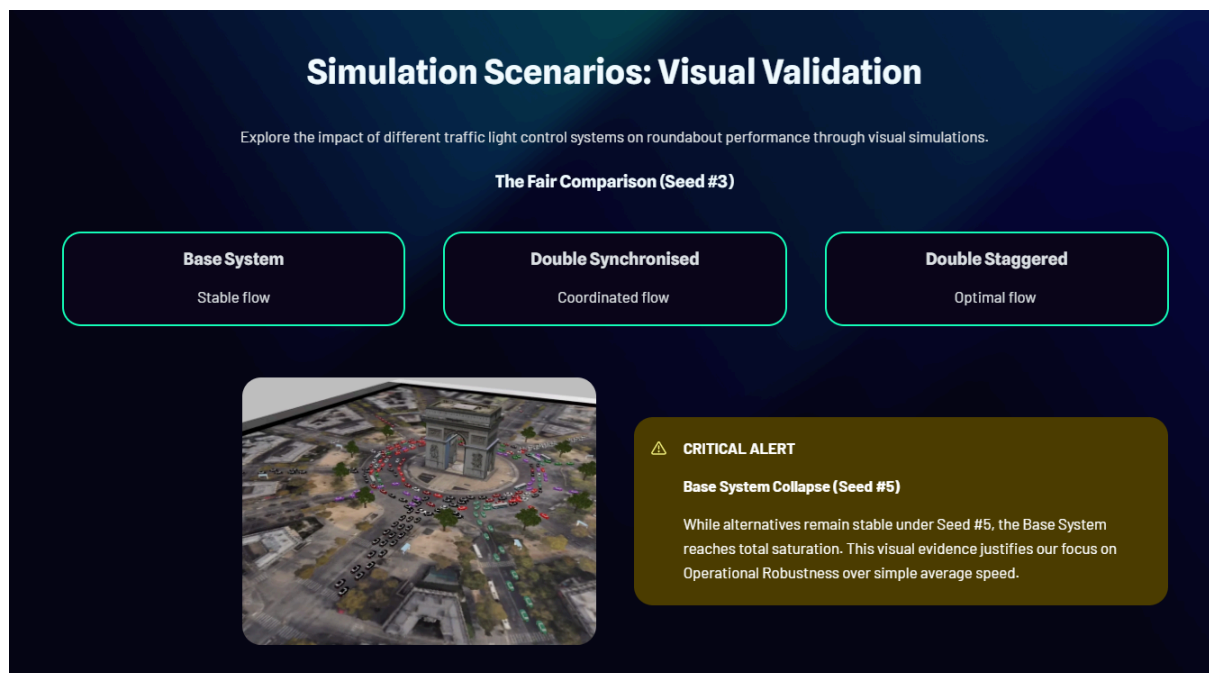
[SEÑALAR BLOQUE DERECHO: DOUBLE STAGGERED]

Finally, we propose the Double Staggered System. In this configuration, the two lights are offset by 5 seconds, with the one furthest from the roundabout turning green first. This manages platoon dispersion, allowing vehicles to find gaps in the circulating flow more efficiently. All systems use a 180-second cycle.

[SEÑALAR TEXTO INFERIOR: FIXED SEEDS & REPLICAS]

We first fixed the random seeds to 1-4 across all scenarios to ensure each system experienced identical event sequences, isolating the effect of the control logic from stochastic variability. We then ran 30 replications per scenario (seeds 1-30) to compute 95% confidence intervals and support statistically robust recommendations.

Diapositiva 11 - 173 palabras (1,6 minutos) - **NAHUEL**



Before diving into the statistical results, we want to provide a visual validation of the system's behaviour. We executed all configurations using the same stochastic sequence: Seed #3.

[REPRODUCIR VIDEOS DE LA FILA SUPERIOR (SEED #3)]

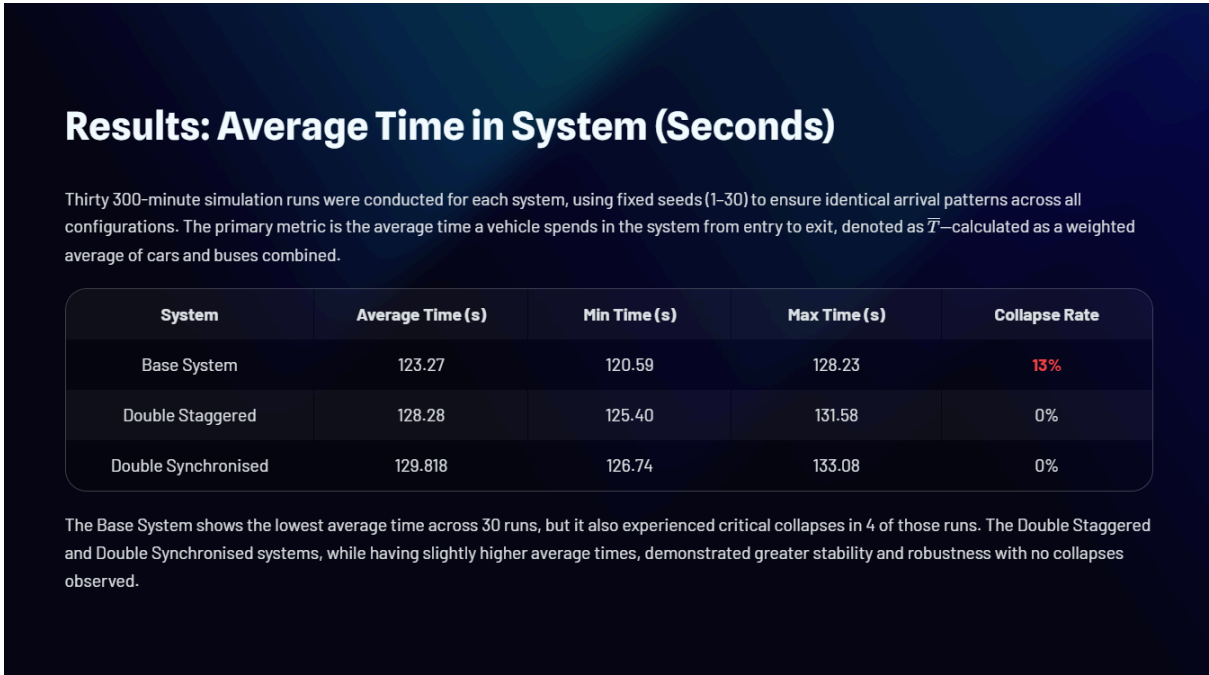
A Seed determines the exact "DNA" of the traffic—meaning vehicle speed and arrival times are identical across these three videos. This reproducibility allows us to conduct a 'controlled experiment' where the only variable changing is the traffic light logic, effectively isolating our results from any random chance. Under these specific conditions, all three systems remain stable, but you can already observe that the Double Staggered System achieves a more fluid and continuous circulatory flow.

[SEÑALAR RECUADRO AMARILLO: CRITICAL ALERT]

However, speed is not our only metric; we are primarily measuring Operational Robustness. To test the true resilience of the current system, we ran it under Seed #5, which introduces a different pattern of demand that pushes the roundabout closer to its limits.

[REPRODUCIR VIDEO DEL COLAPSO (SEED #5)]

This visual evidence justifies our recommendation: while the baseline might be efficient under average flow, it carries a high risk of systemic failure during peak fluctuations.



Having observed the visual behaviour of each system under identical conditions, we then moved to a full statistical validation. Executing 30 runs per configuration (using fixed seeds from 1 to 30) provided these aggregate results for average time-in-system, $\bar{T}$.

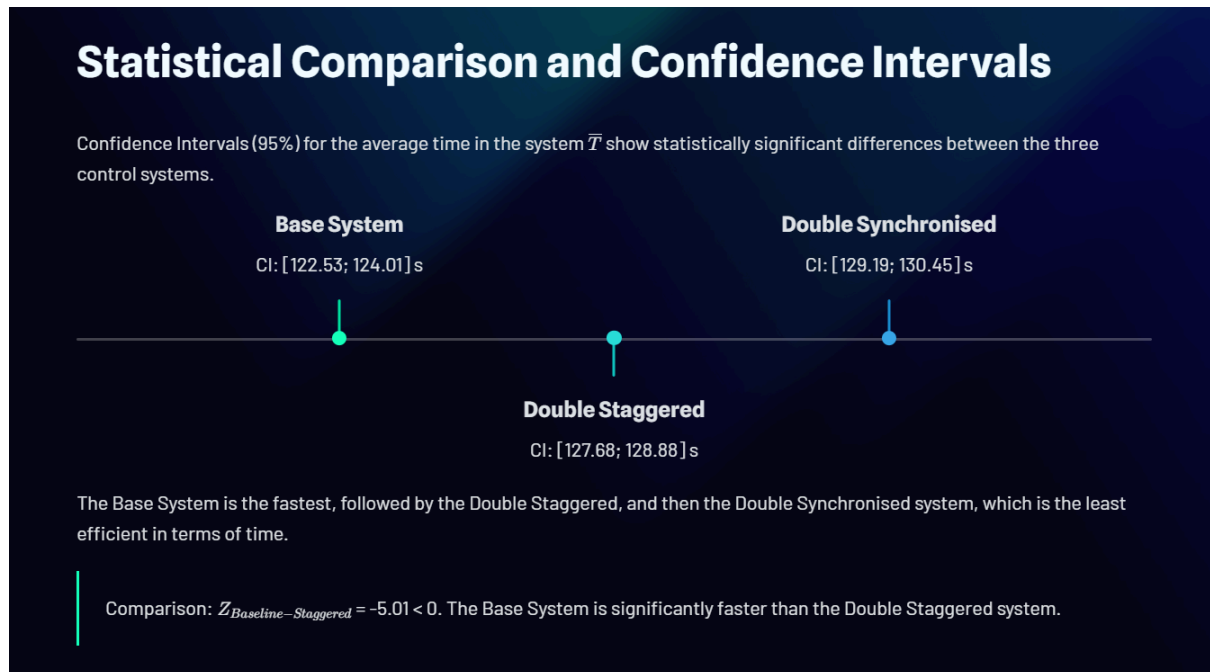
[SEÑALAR FILA: BASE SYSTEM (DESTACAR EL 13% EN ROJO)]

The results indicate a clear trade-off between lower average travel time and operational robustness. As shown in the table, the Base System is the fastest at 123.27 seconds, but it suffered a 13% collapse rate. This means that a non-negligible share of our simulation runs reached a critical state where traffic flow simply broke down.

[SEÑALAR FILAS: DOUBLE STAGGERED Y DOUBLE SYNCHRONISED]

By contrast, both dual-signal configurations completely eliminate collapses. The Double Staggered averages 128.28 seconds, providing a superior balance with a tighter range than the Double Synchronised version at 129.818. Although these averages are approximately 5 seconds higher than the baseline, the total absence of collapses suggests that our proposed signal arrangements prioritise stability. We are effectively exchanging a modest delay for substantially improved reliability.

Diapositiva 13 - 205 palabras (1,9 minutos) - JUAMPI



The confidence intervals at a 95% level provide a rigorous statistical foundation for our comparative analysis.

[SEÑALAR LÍNEA DE TIEMPO / PUNTOS DE COLORES]

As you can see on the timeline, there is no overlap between the intervals of the three control systems. For the Base System, the interval for the weighted average time-in-system $\bar{T}$ is between 122.53 and 124.01 seconds. For the Double Staggered, it is 127.68 to 128.88, and for the Double Synchronised, it ranges from 129.19 to 130.45 seconds.

The clear separation between these non-overlapping intervals confirms that the differences in performance are statistically significant and not just an artefact of stochastic variation between runs.

[SEÑALAR ANCHO DE LOS INTERVALOS]

Beyond significance, the narrow width of these intervals further demonstrates the high precision of our estimated means under the same experimental conditions. This lower variability makes the system more predictable and reliable under high-demand conditions. This allows for an unambiguous ordering: while the Base System is the fastest, the Double Synchronised is the least efficient in terms of travel time.

[SEÑALAR CÁLCULO DE Z AL FINAL]

The Z-score of -5.01 shown at the bottom reinforces that the Base System is significantly faster. However, this 5-second gap is the "statistical price" for the operational robustness and the elimination of system collapses.

Diapositiva 14 - 166 palabras (1,5 minutos) - GUSTAVO

Conclusion: Stability vs. Speed

While the Base System is fastest, it lacks robustness and can lead to critical, unrecoverable congestion. The double traffic light systems stabilize circulation.

System	Description
Base System	Lowest average time, but prone to critical collapse under certain conditions (lacks robustness).
Double Synchronised	Highest average time; stabilizes flow but increases overall delay for all vehicles.
Double Staggered (Recommended)	Best balance: avoids massive bottlenecks, reduces bus travel times (due to better stop integration), and maintains stability.

Recommendation: Prioritising system robustness and better public transport performance, the **Double Staggered system is recommended**, accepting a slight increase in overall average time.

To conclude, our analysis reveals a fundamental trade-off between Operational Stability and nominal Travel Time.

[SEÑALAR BLOQUE SUPERIOR: BASE SYSTEM]

The Base System is indeed the fastest, but it lacks robustness and is the only scenario that suffers a thirteen percent collapse rate, making it an unreliable choice under high-demand peaks in a city like Paris.

[SEÑALAR BLOQUE DERECHO: DOUBLE SYNCHRONISED]

On the other hand, the Double Synchronised System provides the stability the Base System lacks, but it introduces the highest overall delay for all vehicles.

[SEÑALAR BLOQUE INFERIOR: DOUBLE STAGGERED (RECOMMENDED)]

Therefore, we recommend the Double Staggered System as the optimal balance for the Place Charles de Gaulle roundabout. Its five-second offset successfully mitigates bottlenecks and integrates bus dwell times more effectively. Crucially, it also delivers a slightly lower travel time than the Double Synchronised alternative.

[SEÑALAR TEXTO INFERIOR: RECOMMENDATION]

In summary, our recommendation prioritises system robustness and predictability over marginal reductions in average time-in-system. This strategy ensures that the roundabout remains functional and reliable, providing a superior service level for both private vehicles and public transport during peak fluctuations.

Summary & Outlook

Our simulation of the Place Charles de Gaulle roundabout has highlighted critical insights into traffic management strategies.

Base System: Speed vs. Risk

While offering the fastest average travel times, the existing Base System exhibits a significant collapse rate, indicating a lack of robustness under certain conditions.

Double Staggered: Balanced Solution

The Double Staggered traffic light system provides the best balance between efficiency and stability, eliminating collapses and improving public transport integration.

Future Directions

Further research could explore dynamic traffic light adjustments based on real-time flow and their impact on roundabout efficiency.

This study provides a strong basis for implementing more resilient and efficient traffic control systems in complex urban environments.

Moving to a broader perspective, this study provides critical insights into long-term traffic management strategies.

[SEÑALAR BLOQUE IZQUIERDO: BASE SYSTEM: SPEED VS. RISK]

Our findings quantify a fundamental trade-off: while the current configuration is fast, its high collapse rate reveals an operational fragility that simulation effectively identified. From an operational perspective, a system that fails can cause massive disruption, even if its average performance seems strong on paper.

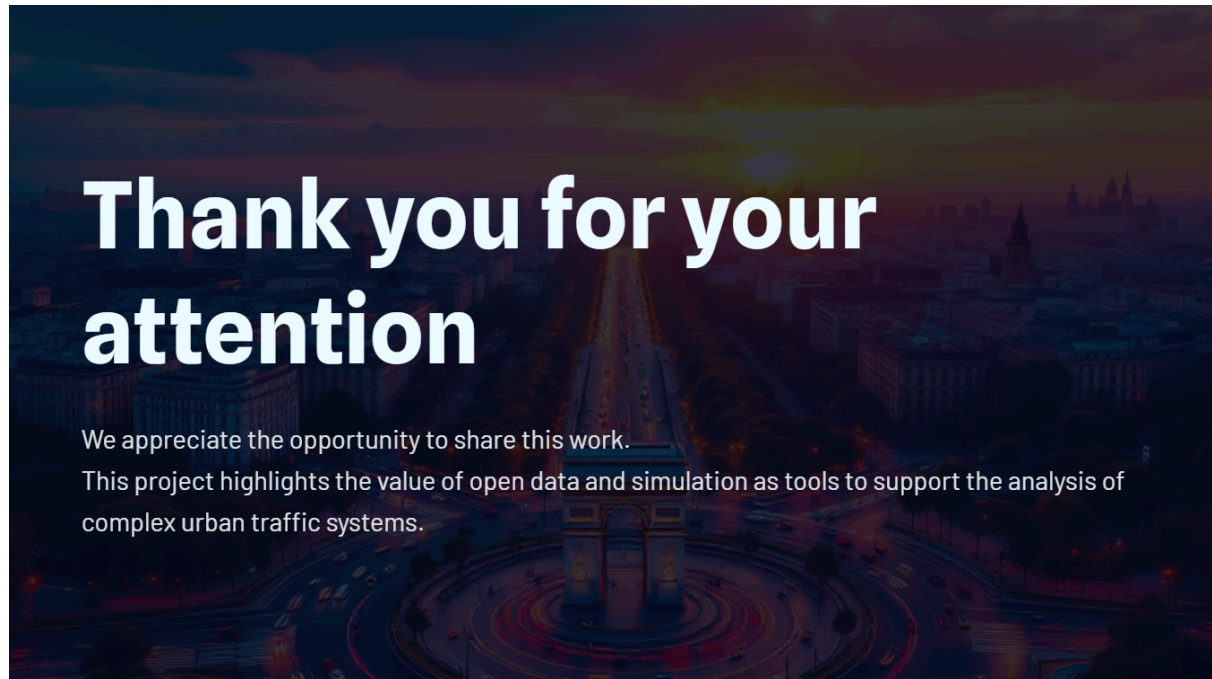
[SEÑALAR BLOQUE CENTRAL: DOUBLE STAGGERED: BALANCED SOLUTION]

In contrast, the Double Staggered System serves as a balanced solution, eliminating collapses while improving public transport integration through a more predictable performance. Methodologically, this underscores that simulation provides a risk-free environment to evaluate evidence-based interventions before any physical implementation.

[SEÑALAR BLOQUE DERECHO: FUTURE DIRECTIONS]

Looking ahead, a natural extension of this research would involve adaptive timing, where signal phases respond to changes in real-time traffic flow, to further optimize efficiency without losing stability.

Ultimately, this work establishes a robust foundation for implementing resilient traffic systems in complex urban environments.



Thank you very much for your attention. It has been a pleasure to present our findings to you today.

We truly appreciate the opportunity to share this work. This project not only proposes an optimized configuration for the Place Charles de Gaulle roundabout but also underscores the immense value of open data and simulation as essential tools for addressing the complexities of modern urban mobility. By using these tools, we can move from intuition-based decisions to evidence-based strategies.

We are now open to any questions regarding the model, the statistical analysis, or our specific findings. In case you are interested in the technical details, we have prepared additional slides covering our full calculation process, as well as a final validation showing that our results remain consistent when scaled from 30 to 100 simulation runs. This reinforces the statistical power of our conclusions and confirms the exceptional stability of the recommended Double Staggered System even under extended testing conditions.

Thank you once again.

Cantidad Total de Palabras de la Parte Obligatoria: 2183

Palabras por Minuto aproximadas: 110

Duración de la Parte Obligatoria de la Presentación (sin contar videos): 20 minutos

PALABRAS

MATEO: 309 (2,9 minutos)

ALEXIS: 254 (2,3 minutos)

JUAN CRUZ: 399 (3,6 minutos)

NAHUEL: 370 (3,4 minutos)

JUAMPI: 371 (3,4 minutos)

GUSTAVO: 316 (2,9 minutos) + Agradecimientos

Results and analysis

Diapositiva 17 - **138 palabras (1,3 minutos)**

Results Analysis

Standard Deviation

Standard deviation measures the average dispersion of values in a dataset relative to its mean. A low standard deviation indicates that data points tend to be close to the dataset's mean, while a high standard deviation indicates that data points are more dispersed.

$$S = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}}$$

Where:

- S : Sample Standard Deviation
- X_i : Each individual value in the dataset
- $\bar{X}$: The mean (average) of the dataset
- n : Sample size

To elaborate on the statistical methodology behind our results, we must highlight that our analysis relies on the sample standard deviation, denoted as S . This metric quantifies the average dispersion of our stochastic simulation outputs around their mean, $\bar{X}$.

[SEÑALAR LA FÓRMULA EN LA DIAPOSITIVA]

By executing 30 independent replications per scenario, we established a sample size that is robust enough to invoke the Central Limit Theorem. This theorem is a cornerstone of our study; it allows us to assume that the distribution of the sample mean is approximately normal, regardless of the underlying distribution of the individual simulation runs.

This provides the solid mathematical foundation required to construct the 95% confidence intervals shown in the previous slides. It ensures that our comparative analysis is not based on "noise" or random chance, but on a statistically rigorous estimation of the system's true performance.

Diapositiva 18 - 165 palabras (1,5 minutos)

Margin of Error

The margin of error is a statistic that expresses the amount of random sampling error in the results of a simulation. It is a measure of random error and expresses the range within which the true value of a population is expected to lie.

$$d = z_{\alpha/2} \cdot \frac{S}{\sqrt{n}}$$

Where:

- d : Margin of Error
- $z_{\alpha/2}$: The critical value from the standard normal distribution (e.g., 1.96 for a 95% confidence level)
- S : Sample Standard Deviation
- n : Sample size

This statistical approach allows us to present results with a high degree of confidence, supporting our recommendations for traffic flow optimization.

Since the true population parameters are unknown, we estimate the range of our results using the Margin of Error, denoted as d . This statistic quantifies the random sampling error and defines the confidence interval within which the true value of the population is expected to lie.

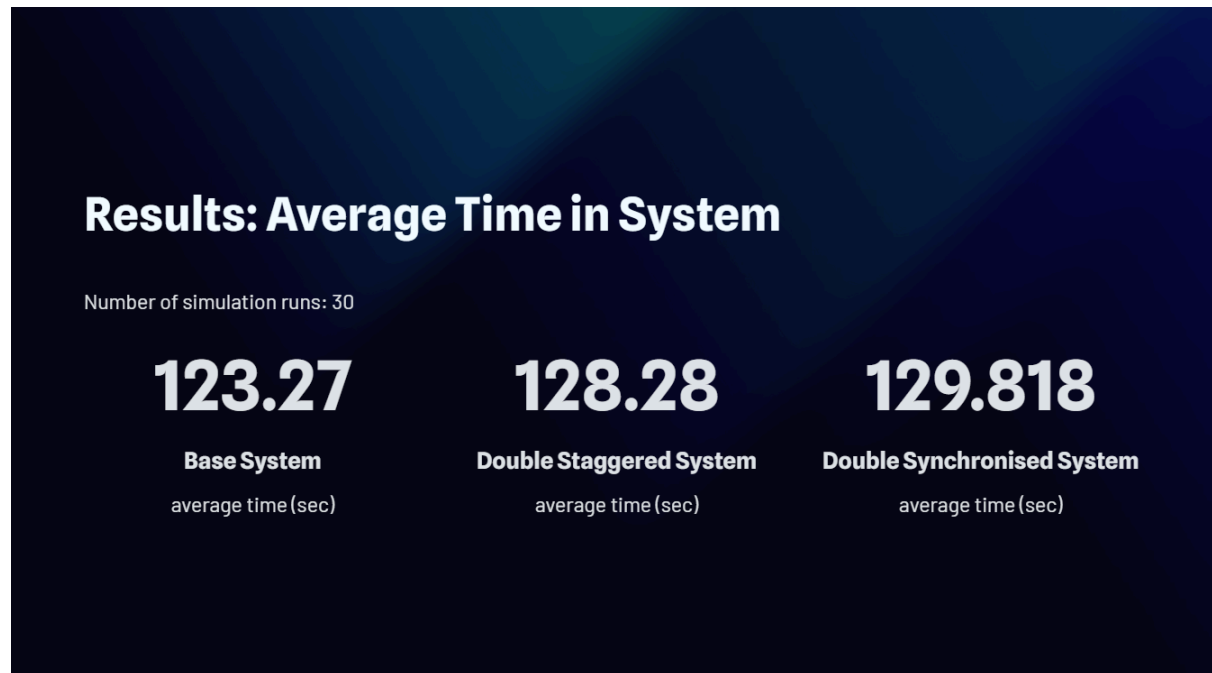
[SEÑALAR LA FÓRMULA]

The margin of error is computed using the expression shown here, where we apply a critical value of 1.96 for our 95% confidence level.

This formula is key to our methodology because it highlights the inverse relationship between the sample size and the error: as we increase the number of replications (n), the margin of error decreases, and our estimates become significantly more precise.

This is exactly why we decided to validate our results with a complementary study of 100 runs. We will show you those results in our final slide to demonstrate that even with a larger sample and a tighter margin of error, our conclusions for the Place Charles de Gaulle roundabout remain stable and technically sound.

Diapositiva 19 - 113 palabras (1 minuto)

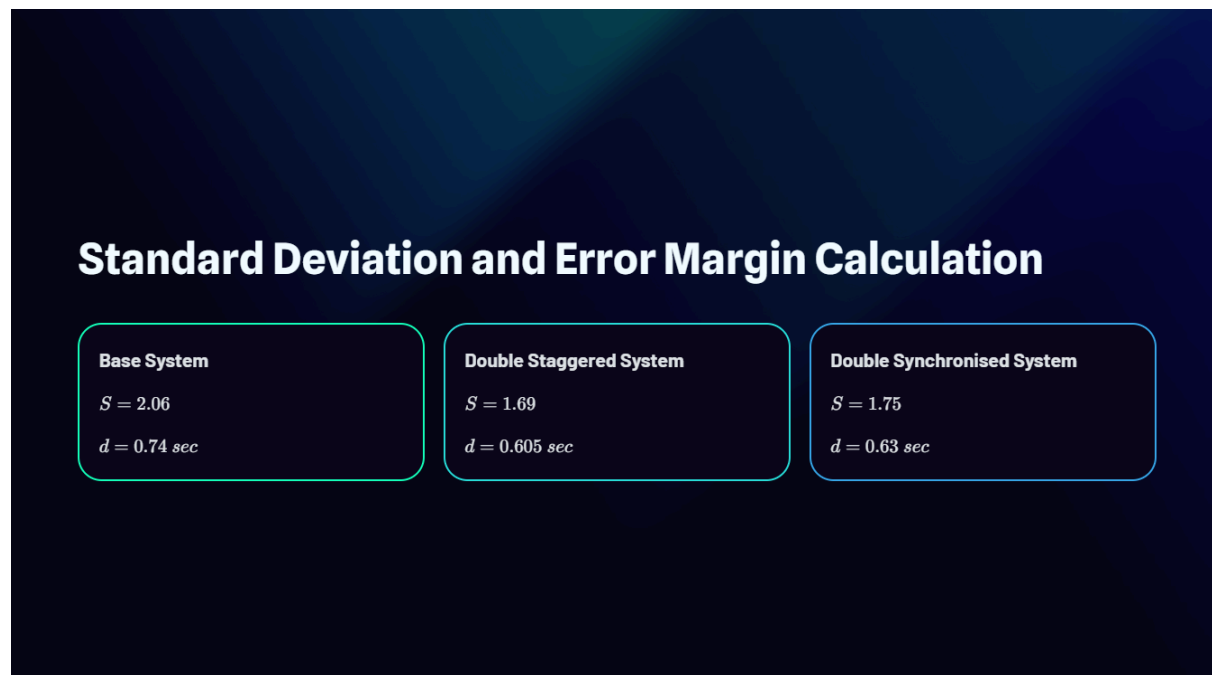


To ground our statistical discussion back into operational reality, let us revisit the point estimates for the average time-in-system, $\bar{T}$, obtained from our initial set of 30 replications.

[SEÑALAR LOS VALORES DE IZQUIERDA A DERECHA]

To recap, the Base System remains the fastest with a nominal value of 123.27 seconds. This is followed by our recommended Double Staggered System at 128.28, and finally, the Double Synchronised System at 129.818 seconds.

While these averages provide a useful starting point for comparison, they only tell part of the story. To determine if these differences are truly significant or simply a result of stochastic variability, we must look beyond the means and analyze the dispersion within our sample.



Applying the mathematical framework we just established to our 30 simulation replications, we obtained these specific metrics for each configuration.

[SEÑALAR BLOQUE CENTRAL: DOUBLE STAGGERED SYSTEM]

As demonstrated here, our recommended Double Staggered System shows the lowest sample standard deviation at 1.69, which results in the narrowest margin of error at 0.605 seconds. This indicates that the results for this configuration are the most consistent across different stochastic scenarios.

[SEÑALAR BLOQUE IZQUIERDO: BASE SYSTEM]

In contrast, the Base System exhibits the highest dispersion, with a standard deviation of 2.06 and a margin of error of 0.74 seconds. The higher variability further highlights its lack of predictability and robustness compared to the signaled alternatives.

[SEÑALAR BLOQUE DERECHO: DOUBLE SYNCHRONISED SYSTEM]

The Double Synchronised System sits in the middle, with a standard deviation of 1.75 and a margin of error of 0.63 seconds.

Overall, these figures confirm that introducing dual-signal control doesn't just manage traffic; it effectively stabilizes the system's performance, reducing the "noise" caused by random vehicle arrivals and bus dwell times.

Diapositiva 21 - 224 palabras (2 minutos)



Finally, to rigorously validate the comparative performance of these systems, we performed a pairwise results comparison using a Z-score analysis. By calculating the Z value for each pair, we confirmed that the observed differences are statistically significant and represent clear operational trends.

[SEÑALAR LAS DOS PRIMERAS COMPARACIONES: BASE VS. SINGALED]

The results clearly indicate a Z-value of minus five point zero one when comparing the Base and Staggered systems, and minus six point five for the Synchronised one. Since both values are significantly below the zero threshold, it confirms that the Base System is indeed the fastest in terms of nominal travel time. However, as we have established, this speed comes at the cost of a high risk of systemic collapse.

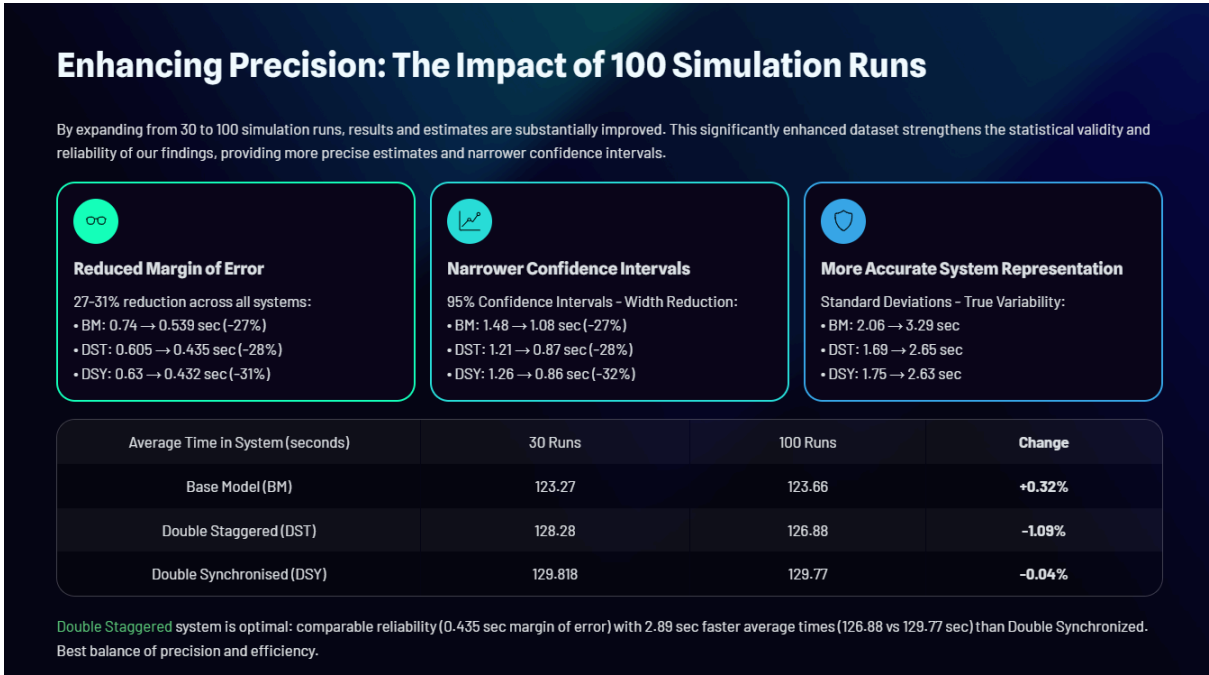
[SEÑALAR LA TERCERA COMPARACIÓN: STAGGERED VS. SYNCHRONISED]

The critical finding for our recommendation lies in the third comparison: a Z-score of minus one point five between the two dual-singaled options. This magnitude confirms that the Double Staggered system is significantly more efficient than the Double Synchronised one.

This reinforces our proposal: the Double Staggered configuration offers the optimal balance. It provides the operational stability that the Base System lacks while delivering a superior efficiency compared to the Double Synchronised option.

To conclude this technical block, we will now look at our final slide, where we demonstrate that these exact trends and rankings remain remarkably stable even when scaling the experiment to 100 simulation runs.

Diapositiva 22 - 241 palabras (2,2 minutos)



To culminate our technical validation, we scaled our experimental design from 30 to 100 simulation replications. This expansion significantly bolsters the statistical power of our findings by providing narrower confidence intervals and a substantial reduction in sampling margin of error.

[SEÑALAR RECUADRO IZQUIERDO: REDUCED MARGIN OF ERROR]

As you can see, the margin of error, d , was reduced by up to 31% across all systems. Specifically for our proposed Double Staggered System, the error margin dropped to a remarkable 0.435 seconds, ensuring a highly precise estimation of its operational behaviour.

[SEÑALAR RECUADRO DERECHO: MORE ACCURATE SYSTEM REPRESENTATION]

It is important to note the increase in standard deviations, S , from the 30-run sample. This is a positive indicator of stochastic convergence; the larger sample is more effectively capturing the true variability of the system, including low-probability traffic fluctuations that smaller samples might miss.

[SEÑALAR TABLA INFERIOR]

The most vital takeaway is that despite the increased precision, our rankings remain unchanged. In fact, the performance gap between the two signaled alternatives has widened significantly.

While the initial difference with 30 runs was 1.538 seconds, with 100 runs this gap nearly doubled to 2.89 seconds—an 87.9% increase in the Double Staggered's lead over the Double Synchronised option. This confirms that as we achieve stochastic convergence, the advantages of the Double Staggered System become even more pronounced.

In light of this evidence, our model is stable and precise, providing a robust foundation for the traffic management strategy we are proposing for the Place Charles de Gaulle roundabout.

Cantidad Total de Palabras de la Parte Opcional de Análisis de Resultados: 1038

Palabras por Minuto aproximadas: 110

Duración de la Parte Opcional de Análisis de Resultados: 9,5 minutos

MEJORAS

- En el estudio de las 30 corridas reemplazar el tiempo promedio por un número muy alto para la escala cada vez que haya un colapso, tal número puede ser 500s o algo así.
- La prioridad de entrada evita colas sobre las avenidas, la de salida puede formarlas, pero descongestiona la rotonda. Se podría plantear una señalización que cambie la prioridad dependiendo el tráfico o el estudio de tal escenario.
- Realizar un estudio de costos de implementación del sistema desfasado recomendado ("Double Staggered"), y compararlo con el sistema base y el sincronizado.